

Quiz 2: Probability and Statistics (30 mks)

Each question: 5 marks

1. Suppose $Z = X + U$ where X is exponential random variable with parameter λ and U is Uniform $[0, 1]$ variable. Derive the first two moments of Z using MGF. (You have to derive MGF of X and U first).

2. Let $X_1, X_2, X_3, \dots$ be a sequence of Laplacian (doubly exponential) continuous random variables such that

$$f_{X_n}(x) = 0.5ne^{-n|x|}, \quad x \in \mathbb{R}.$$

Show that X_n converges in probability to 0.

3. Give me a procedure to convert samples from an Exponential random variable to *Uniform* $[0, 1]$. Justify why the procedure is correct.

4. Let (X_1, X_2) be a continuous random vector with joint density

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 1, & 0 < x_1 < 1, 0 < x_2 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation

$$Y_1 = X_1 - X_2, \quad Y_2 = X_1 + X_2.$$

Obtain the joint density $f_{Y_1, Y_2}(y_1, y_2)$ in terms of $f_{X_1, X_2}(x_1, x_2)$.

10 marks

Let $X_n \sim \text{Exp}(n)$, i.e. its density is

$$f_{X_n}(x) = ne^{-nx}, \quad x \geq 0.$$

Define $Y_n = nX_n$.

1. Show that Y_n converge in distribution to $Y = \text{Exp}(1)$.
2. Does Y_n converge in probability to Y ? Justify.
3. Does Y_n converge almost surely to Y ? Justify.
4. Does Y_n converge in mean (i.e. in L^1)? Justify.